[Layer 3: Planning Layer (Orientation Strategy)]

Start

Retrieve tracked sheet pose
(x, y, z, Rz, RY, Rx)

Retrieve zone-based admissibility window
[RZ_min, RZ_max]

Is tool Z-axis pointing downward?
cos(RX) * cos(RY) < -0.10

No

Preprocessing: enforce tool Z-axis downward (apply 180° flip if needed)

Yes

LEVEL 1

Apply reference-face normalisation

Compute ZYX Euler angles (RZ, RY, RX)

Is RZ within zone admissibility window [RZ_min, RZ_max]?

Select reference-face pose

Admissibility confirmed

LEVEL 2

Evaluate detected-pose orientation

Is RZ within zone admissibility window [RZ_min, RZ_max]?

Select detected pose

Admissibility confirmed

Compute inverted orientation
(RZ ± 180°, RY = –RY, RX = –RX)

Mark admissibility unconfirmed and propagate flag

Forward selected orientation to:
Grip Planning (Layer 3)

Motion Planning (Layer 4)